

atm_cover.vr

File location: \$VCS_HOME/doc/examples/nativetestbench/openvera/rvm/atm/atm_cover.vr

```
//
// ATM cell functional coverage model
//
class atm_cover {
    protected atm_cell cell;
    protected event cover_it;
    coverage_group atm_header {
        sample_event = sync(ALL, this.cover_it) async;
        sample GFC (cell.gfc);
        sample VPI (cell.vpi) {
            state zero (8'h00);
            state ones (8'hFF);
            wildcard state walk1_0 (8'bxxxxxxx1) if (cell.vpi != 8'hFF);
            wildcard state walk1_1 (8'bxxxxxxx1x) if (cell.vpi != 8'hFF);
            wildcard state walk1_2 (8'bxxxxxx1xx) if (cell.vpi != 8'hFF);
            wildcard state walk1_3 (8'bxxxxx1xxx) if (cell.vpi != 8'hFF);
            wildcard state walk1_4 (8'bxxxx1xxxx) if (cell.vpi != 8'hFF);
            wildcard state walk1_5 (8'bxxx1xxxxx) if (cell.vpi != 8'hFF);
            wildcard state walk1_6 (8'bx1xxxxxx) if (cell.vpi != 8'hFF);
            wildcard state walk1_7 (8'b1xxxxxxx) if (cell.vpi != 8'hFF);
            wildcard state walk0_0 (8'bxxxxxxx0) if (cell.vpi != 8'h00);
            wildcard state walk0_1 (8'bxxxxxx0x) if (cell.vpi != 8'h00);
            wildcard state walk0_2 (8'bxxxxx0xx) if (cell.vpi != 8'h00);
            wildcard state walk0_3 (8'bxxxx0xxx) if (cell.vpi != 8'h00);
            wildcard state walk0_4 (8'bxxx0xxxx) if (cell.vpi != 8'h00);
            wildcard state walk0_5 (8'bx0xxxxxx) if (cell.vpi != 8'h00);
            wildcard state walk0_6 (8'bx0xxxxxx) if (cell.vpi != 8'h00);
            wildcard state walk0_7 (8'b0xxxxxxx) if (cell.vpi != 8'h00);
        }
        sample CLP (cell.clp);
        sample PT (cell.pt);
        sample HEC (cell.hec) {
            state good (0);
            state bad (1:255);
            wildcard state walk1_0 (8'bxxxxxxx1);
            wildcard state walk1_1 (8'bxxxxxxx1x);
            wildcard state walk1_2 (8'bxxxxxx1xx);
        }
    }
}
```

```
        wildcard state walk1_3 (8'bxxxx1xxx);
        wildcard state walk1_4 (8'bxxx1xxxx);
        wildcard state walk1_5 (8'bx1xxxxxx);
        wildcard state walk1_6 (8'bx1xxxxxxx);
        wildcard state walk1_7 (8'b1xxxxxxx);
    }
}

task cover(atm_cell cell) {
    this.cell = cell;
    trigger(this.cover_it);
}
}
```